



BITS Pilani

Pilani Campus

CE F230

Civil Engineering Materials (Lab)

Instructor-in-charge : Dr. Mukund Lahoti

Instructors : Mr. Anuj Adhikari, Mr. Gulshan Kumar Yadav,
Mr. Potdar Namdev Mukund , Ms. Ankita Pareek,
Mr. Thakur Ramjiram Singh



List of Experiments

Experiment No.	Name of experiment
	Introduction to all experiments, marking scheme, lab report format, etc.
1.	Brick: a) Compressive strength, water absorption, efflorescence tests b) Making fly ash bricks
2.	Cement: a) Heat of hydration of cement b) Test brick samples for compressive strength and efflorescence
3.	Cement: a) Normal consistency and Setting time b) Fineness and Specific gravity of cement
4.	Fine aggregate: a) Bulking of sand b) Silt content c) Specific gravity
5.	Fineness modulus and grain size distribution: Fineness modulus and grain size distribution

List of Experiments



Experiment No.	Name of experiment
6.	Cement mortar: a) Cast 3 cubes for compressive strength testing (air dried in shade, water-cured) b) Cast 3 briquettes for tensile strength testing
7.	Concrete (casting): Nominal mix design a) Cast 2 cylinders, 2 cubes, 1 beam b) Test workability (slump)
8.	Concrete (casting): IS 10262 mix design a) Cast 2 cylinders, 2 cubes, 1 beam b) Test workability (slump)
9.	Concrete (testing): a) Modulus of rupture (beam test) b) Test compressive strength of cube and cylinder c) Split tensile strength of cylinder
10.	Demo to various tests on: a) Rapid Chlorine Penetration Test (RCPT) b) Non Destructive Testing (NDT) c) Rebar Testing

Lab Manual



- Gambhir, ML and Neha Jamwal, Lab Manual- Building and Construction Materials (Testing and quality control), Mc Graw Hill, 1st edition 2014
- Moondra, HS and Rajiv Gupta, Lab Manual for Civil Engineering, CBS 2nd edition 2000

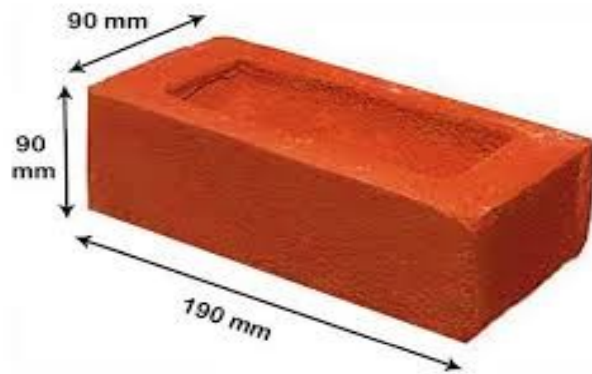
Construction Materials



- Stone
- Brick
- Lime
- Cement
- Sand
- Aggregate
- Concrete
- Steel
- Timber
- Bitumen



Experiment 1: Bricks



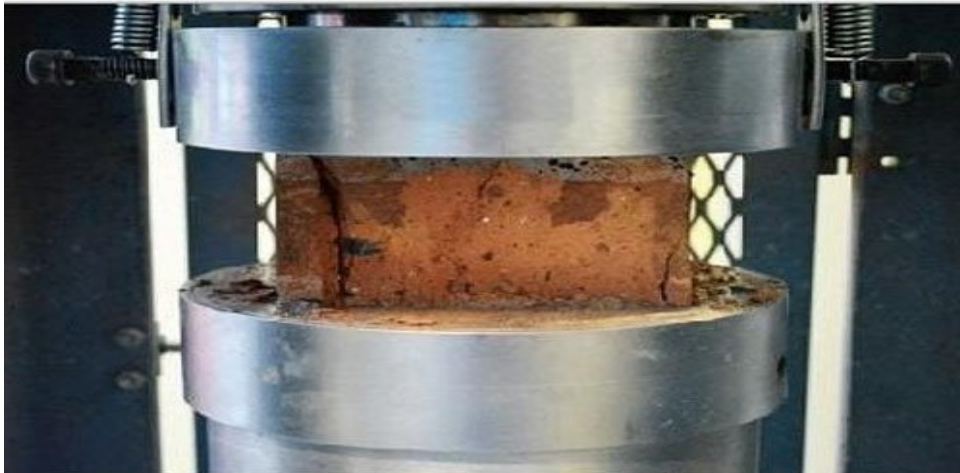
Tests to be carried out in Laboratory

- (a) Compressive Strength & Efflorescence Tests
- (b) Making fly ash bricks



1(a): Compressive strength

- Test the bricks for compressive strength
- Check which class it belongs to



(IS:1077-1992)



Compression testing machine

Efflorescence tests



- White patches occurs on the surface of the bricks **due to deposit of salts**.
- Presence of sulphate, sodium, potassium will cause efflorescence.
- **Flaking** will be occurred in serious efflorescence
- **Classification:** Slight, moderate, heavy, serious



Test Setup

(IS:3495-1992)

1(b): Making Fly Ash Bricks



- Some advantages of fly ash bricks over conventional bricks:
 - ✓ High Fire Insulation,
 - ✓ Easy to transport ,
 - ✓ Lower water absorption,
 - ✓ Do not require soaking in water



Ingredients of fly ash bricks

Fly ash

Lime Sludge

Stone Dust

Gypsum



Experiments 2 and 3: Cement



Tests to be carried out in Laboratory

2: Heat of hydration of cement

3(a): Normal Consistency & Setting Time

3(b): Fineness of Cement



Types of Cement

Experiment 2: Heat of hydration of Cement



- Hydration of cement is a exothermic reaction (approx. 120 cal/g).
- Temperature will be increased in hydration of cement.



Heat of Hydration Machine

(IS 11262 -1985)

Experiment 2(b): Test brick samples for compressive strength and efflorescence



Compression testing of brick with mortar



Compression testing machine



Measurement of efflorescence in bricks

Experiment 3(a): Normal Consistency of cement

- The amount of water required to make the cement paste of standard consistency
- Degree of wetness of cement.

(IS: 4031 (Part 4) – 1988)

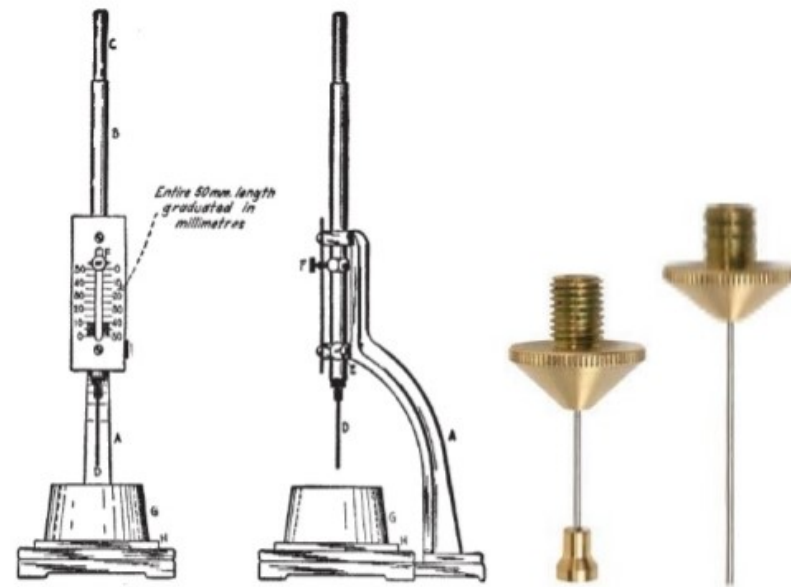


Vicat Apparatus

Setting Time of Cement



Vicat Apparatus



Needles

(IS: 4031 Part 5 – 1988)

Experiment 3(b): Fineness of Cement



- 90 micron IS sieve are used



Sieve Lid



Weighing Balance



Sieve with Pan



Sieve Shaking Machine

(IS: 4031 Part 1 – 1988)

Experiment 4: Fine aggregate



- Aggregates that pass through 4.75 mm sieve
- Widely used in construction industry to increase the volume of concrete, thus it is a cost saving material

Tests to be carried out in Laboratory

- a) Bulking of Sand
- b) Silt Content
- c) Specific Gravity
- d) Water Absorption



River sand



Manufactured sand

Experiment 4(a): Bulking of Sand

- Volume of dry **sand** increases due to absorption of moisture.
- Volume increase of dry **sand** is known as **bulking of sand**.

3rd Trail



4th Trail

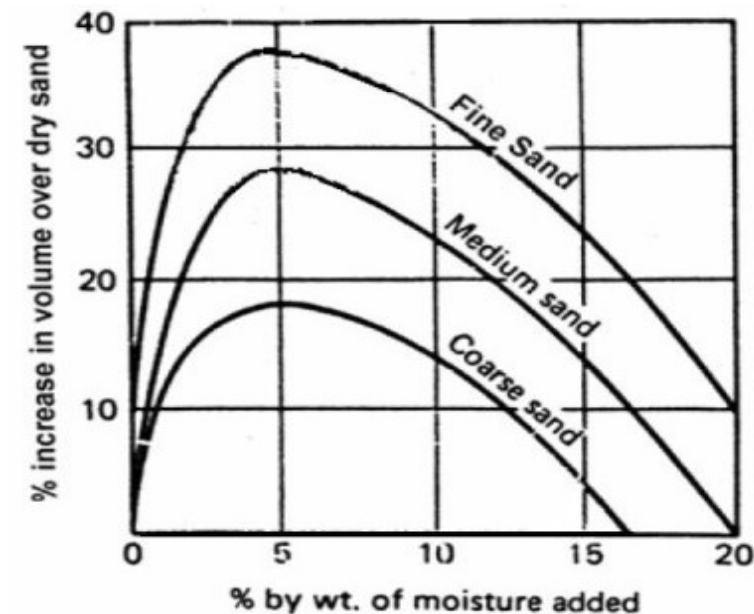


5th Trail



(IS 386 part 3-1963)

Test Set up



Experiment 4(b): Silt Content



- Silt content is fine material (less than 75 micron)
- Reduce the strength and causes rework
- Check whether it satisfies allowable limit



Clay layer – water clears

Silt layer – 2 hours

Sand layers – 1 minute

(IS 2386-2 1963)

Silt Content in Fine Aggregate

Experiment 4(c): Specific Gravity of sand



- Important to know the specific gravity to accurately batch sand for making concrete
- We use Pycnometer method



Pycnometer

(IS:2386 Part 3 -1963)

Experiment 5: Fineness modulus and grain size distribution



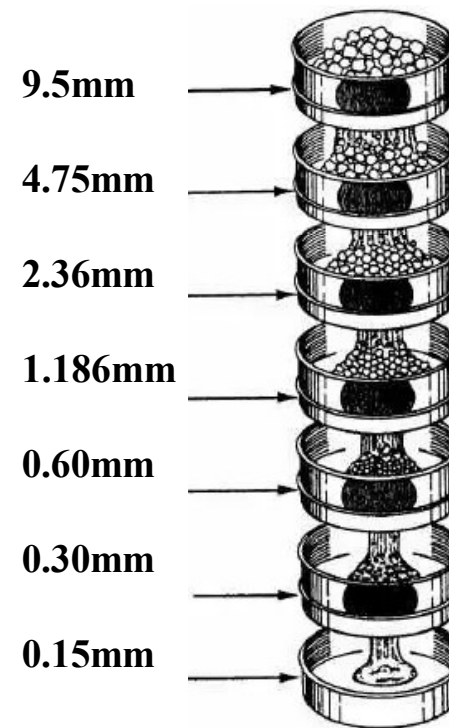
Fineness modulus and grain size distribution

TABLE 5.7 Sample Calculation of Fineness Modulus

Sieve Size	Percentage of Individual Fraction Retained, by Weight	Cumulative Percentage Retained by Weight	Percentage Passing by Weight
9.5 mm (3/8 in.)	0	0	100
4.75 mm (No. 4)	2	2	98
2.36 mm (No. 8)	13	15	85
1.18 mm (No. 16)	25	40	60
0.60 mm (No. 30)	15	55	45
0.30 mm (No. 50)	22	77	23
0.15 mm (No. 100)	20	97	3
Pan	3	100	0
Total	100	$\Sigma = 286$	

$$\text{Fineness Modulus} = 286/100 = 2.86$$

(IS:383-1970)



Experiment 6: Mortar



- Mortar used in masonry construction to fill gaps between bricks and blocks.
- Mortar is mixture of sand, a binder such as cement or lime, and water



Cement



Sand



Water



Mortar



Experiment 6: Cement mortar casting



- Cast 6 cubes for compressive strength testing (air dried in shade, water-cured)
- Cast 3 briquettes for tensile strength testing



Briquette Mould



Cube Mould

IS:4031 part-6 1988

Experiment 7: Concrete (Nominal mix design)



- Determine the relative proportions to make concrete as **durable** and as **economical** as possible

- Nominal mix design:**

✓ M15 and M20 are the grade of concretes

Concrete Grade	Ratio
M5	1:5:10
M7.5	1:4:8
M10	1:3:6
M15	1:2:4
M20	1:1.5:3

M15

Mix ratio = 1:2:4

1 cement content

2 sand content

4 aggregate content

Cast 2 cylinders,
2 cubes and 1
beam



Cement

= ?



Sand

= ?



Aggregate

= ?



Water

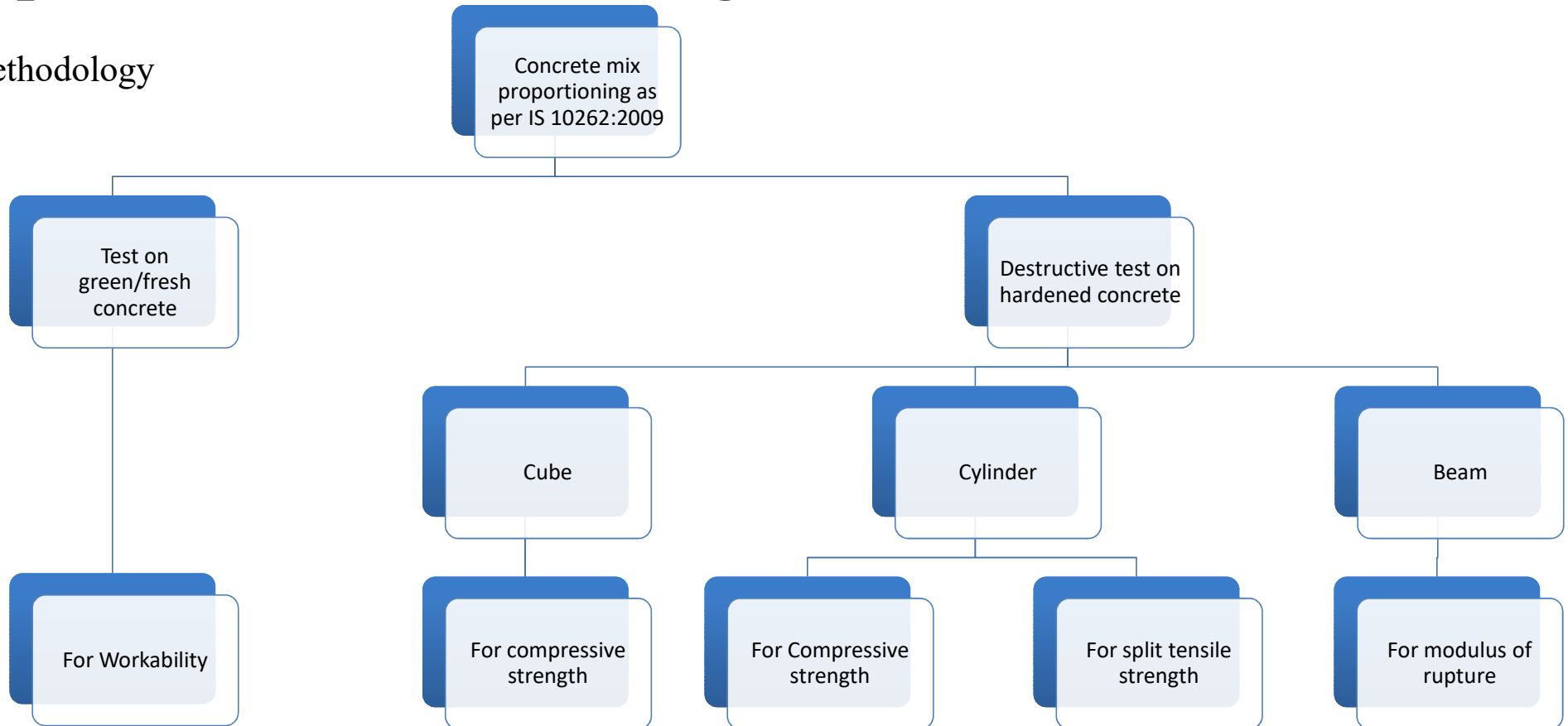
= ?

Experiment 8: Concrete (IS 10262:2009 mix design)

Experiment 9: Concrete testing



Methodology



Experiment 8: Concrete (IS 10262 mix design)



Component	Quantity (kg/m ³)
Cement	350
Water	140
Fine Aggregate	896
Coarse Aggregate	1140
Chemical admixture	7
Water-cement ratio	0.4

Tentative values taken from an example in ANNEX-A IS 10262:2009



Experiment 9: Concrete testing, green/fresh concrete



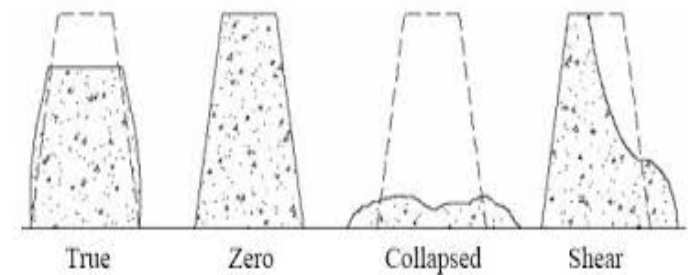
- Slump Test



Procedure



Equipment



Result

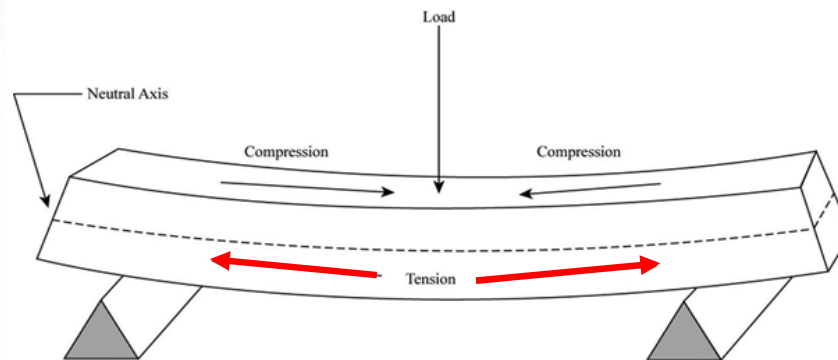
Experiment 9: Concrete testing, hardened concrete



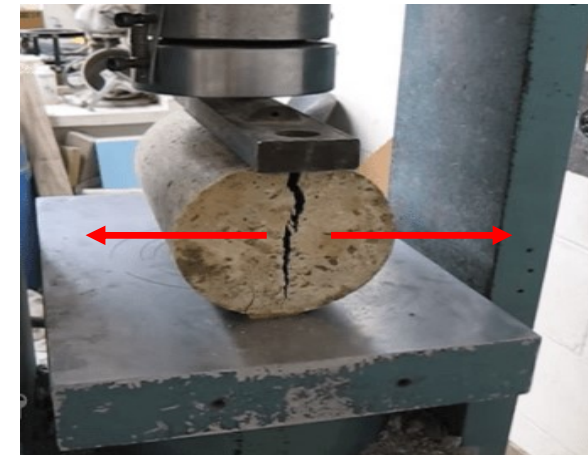
- Modulus of rupture of concrete
- Compressive strength of concrete
- Split tensile strength of concrete



Modulus of rupture test-on beam specimen



Compressive strength test- on cube & cylinder specimens



Split tensile strength test- on cylinder specimen

Experiment 10: Demo



- Rapid Chlorine Penetration Test (RCPT)
- Non-Destructive Testing (NDT)
- Rebar Testing



Electrical indication of concrete's ability to resist chloride ion penetration
Standards - ASTM C1202, AASHTO T277



Rebound hammer test



Rebar test using Universal Testing Machine(UTM)

Key points



- Attend all labs, absence in lab will cost Zero mark for that particular experiment.
- 1 Lab as a buffer will be conducted at the end of the semester, for genuine cases.
- Student will maintain a Lab Record Book, which they must bring every time during Lab hours
- A viva will be conducted at the end of every lab
- A final quiz will be conducted by the end of semester. Quiz will be based on the experiments performed in lab.

Lab Evaluation



Lab Components Breakups:

Lab Record: 25%

Viva : 25%

Final Quiz: 50%



BITS Pilani

Pilani Campus

THANK YOU